

48MM OD INTERNALLY WIRED BAR GRID

Technical data / extra information sheet

Mechanical Theatre Equipment Compliance Portal

Purpose of this sheet

A technical record template for this equipment. Complete and review it using the actual manufacturer instructions, installation design, inspection records and site assessment. It does not certify an asset or assign a load rating.

1. Equipment identification

Field	Details
Equipment name	48mm OD Internally Wired Bar Grid
Equipment category	Lighting Bars
Manufacturer / supplier	To be confirmed for the actual asset.
Model / product variant	To be confirmed; include part number and version.
Asset / serial / batch ID	To be completed; attach a photograph of identification markings.
Site / location	To be completed.
Owner / responsible person	To be completed.
Status / record date	Generic template - asset details and approval outstanding. Date: _____

2. Description and typical use

A grid of nominal 48 mm outside-diameter bars used to support lighting or other approved equipment. This assembly also contains electrical wiring and outlets, requiring both structural and electrical records.

- Use only the documented support layout and approved equipment attachment locations.
- Include fixtures, cabling, adaptors and the assembly's own weight in the load record.

3. Technical data to confirm

Enter confirmed values and the document/drawing reference in the asset record. WLL means working load limit. No numerical load rating is assigned by this sheet.

Parameter	Required confirmation
Section and material	Nominal outside diameter: 48 mm. Verify actual OD, wall thickness, grade and bar material.
Dimensions/layout	Record grid plan, member lengths, intersections and support coordinates.
Design load	Confirm allowable point/distributed loads, deflection and each support reaction.
Joints and supports	Identify couplers, splices, suspension fittings and the supporting structure.
Attachment compatibility	Confirm clamps suit the actual 48 mm OD section, wall and permitted local loading.
Electrical installation	Record circuit/socket ratings, wiring, earthing/bonding, protection, tests and isolation arrangements.
Operating limits	Record all manufacturer/designer restrictions and the approved load arrangement.
Inspection / maintenance	Record pre-use checks, periodic inspections and servicing responsibilities; use the exact product instructions.
Documents / traceability	Attach applicable instructions, drawings, certificates and inspection/examination reports. Record reference and revision.

Important limitation

Structural capacity and electrical capacity are separate limits. Isolate before relevant electrical work; holes, sockets and internal fittings must be included in the structural assessment.

4. Regulatory and good practice context

Topic	Application to record
Work equipment	Apply relevant PUWER duties for suitability, maintenance, inspection and safe use.
Lifting equipment	Assess the actual role in any lifting system. A fixed support or structural component should not be classified solely from its name or portal category.
Standards / instructions	Identify relevant product/structural standards and any BS EN 17206 machinery interface. Confirm applicable editions and scope; use maker/designer limits.
Electrical safety	Apply relevant Electricity at Work Regulations duties. Record a suitable maintenance, inspection and testing regime for the wiring and outlets.

5. Pre-use and periodic inspection

Use competent personnel and a safe inspection method. These checks support the asset-specific regime; they do not replace maintenance or a statutory thorough examination. Record Pass / Fail / N/A and findings.

Check	What to confirm	Result / notes
Identification/records	Asset and configuration identifiable; instructions and previous defect actions available.	
Bar condition	No permanent bending, cracking, damaging dents or significant corrosion.	
Joints/supports	All specified joints, clamps and fixings secure; no rotation or slip.	
Load distribution	Positions and weights match the approved grid load plan.	
Electrical condition	No damaged sockets, exposed conductors or missing covers; electrical checks by a competent person.	
Change or damage history	Check for overload, impact, modification, repair or changed environment since last assessment.	
Safe function check	Where applicable, verify intended operation using an authorised safe procedure; do not improvise a proof-load test.	
Outcome/actions	Record defects, restrictions and actions; confirm who authorises return to use.	

6. Removal from service triggers

- Structural cracking, permanent deformation, insecure support or evidence of overload.
- Electrical damage, failed protective measures or unverified structural/electrical alterations.
- Damage, wear, corrosion or malfunction beyond the manufacturer/designer acceptance criteria, or uncertainty about safe condition.
- Suspected overload, significant impact or an unapproved modification: isolate the affected asset pending competent assessment.
- Tag and prevent use of unsafe equipment; record the defect and authorised repair/replacement and reinspection before return.

Inspection planning

Where LOLER applies, record the correct examination interval or a competent person's written scheme. Do not assume that an annual visual check is sufficient for every asset. Include examinations after relevant exceptional events.

7. Site-specific notes and limitations

Topic	Site entry / supporting document reference
Permitted loads / uses	
Prohibited uses / limits	
Location / environment	
Inspection / servicing plan	
Related assets / drawings	

8. Inspection and document record

Date	Check / document	Completed by	Outcome / action

Asset details verified by: _____ Date: _____

Release / approval reference: _____

Reference guidance

[HSE - maintaining electrical equipment safety](#)

[HSE - PUWER overview](#)

[HSE - LOLER overview](#)

[HSE - thorough examinations](#)

References reviewed 9 October 2026. Example-brand documents do not identify the asset or approve another manufacturer's product. Keep the exact product manual and installation evidence with the portal record.

Portal document note

Suggested portal path: documents/bar_48mm_internally_wired_grid_technical_datasheet.pdf. Uploading the file only makes it available; the equipment record must link to this exact filename.